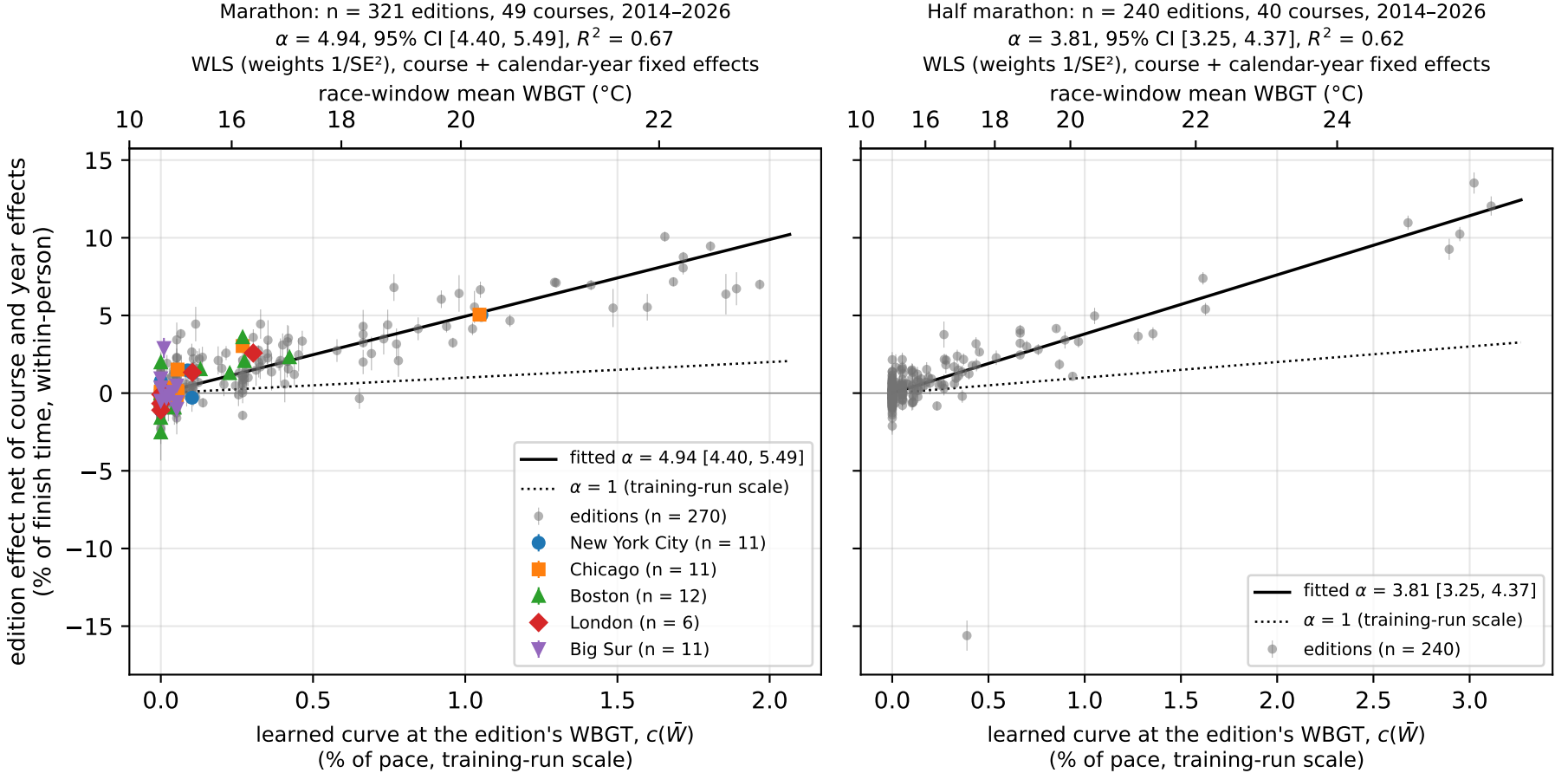
Marathon: n = 321 editions, 49 courses, 2014–2026
α = 4.94, 95% CI [4.40, 5.49], R² = 0.67
WLS (weights 1/SE²), course + calendar-year fixed effects
race-window mean WBGT (°C)
10
16
18
20
22
edition effect net of course and year effects
(% of finish time, within-person)
15
10
5
0
−5
−10
−15
fitted α = 4.94 [4.40, 5.49]
α = 1 (training-run scale)
editions (n = 270)
New York City (n = 11)
Chicago (n = 11)
Boston (n = 12)
London (n = 6)
Big Sur (n = 11)
0.0
0.5
1.0
1.5
2.0
learned curve at the edition's WBGT, c(W̄)
(% of pace, training-run scale)
Half marathon: n = 240 editions, 40 courses, 2014–2026
α = 3.81, 95% CI [3.25, 4.37], R² = 0.62
WLS (weights 1/SE²), course + calendar-year fixed effects
race-window mean WBGT (°C)
10
16
18
20
22
24
fitted α = 3.81 [3.25, 4.37]
α = 1 (training-run scale)
editions (n = 240)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
learned curve at the edition's WBGT, c(W̄)
(% of pace, training-run scale)